



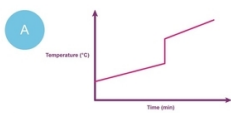
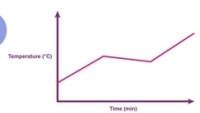
Methods of separation and purification

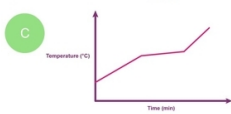
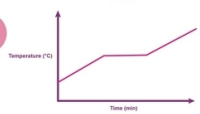
Name:

Date:

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A solid sample of a pure substance is heated until its temperature is above its melting point. The temperature of the sample is recorded and plotted on a graph against time. Which is the correct heating curve?

A  **B** 

C  **D** 

Correct Answer: A B C D

Explanation:

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A 2.0 g sample of potassium chlorate(V) crystals was put into a boiling tube. Then 8.0 g of water was added. The mixture was heated in a hot water bath until all the crystals dissolved. Then the hot solution was stirred with a thermometer as it cooled. The first crystals appeared when the temperature was 63 °C.

What is the solubility of potassium chlorate(V) in water at 63 °C?

A 25 g per 100 g water **B** 50 g per 100 g water

C 40 g per 100 g water **D** 80 g per 100 g water

Correct Answer: A B C D

Explanation:

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The table shows the solubilities of potassium chlorate(V) and potassium chloride in water at 20 °C and at 80 °C. A mixture of 2.0 g potassium chlorate(V) and 2.0 g potassium chloride was warmed to 80 °C with 10 cm³ water until all the crystals dissolved. Then the solution was cooled to 20 °C. What masses of potassium chlorate(V) and potassium chloride crystallised from the solution?

	Solubility/g per 100 g at 20 °C	Solubility/g per 100 g at 80 °C
Potassium chlorate(V)	7.0	27
Potassium chloride	34	51

A 0 g potassium chlorate(V) and 0 g potassium chloride **B** 1.3 g potassium chlorate(V) and 0 g potassium chloride

C 0.7 g potassium chlorate(V) and 1.4 g potassium chloride **D** 2.0 g potassium chlorate(V) and 2.0 g potassium chloride

Correct Answer: A B C D

Explanation:

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A small spot of a solution S was placed on the start line of a sheet of chromatography paper. Reference spots of solutions of four metal ions were also added to the start line. The paper was stood in a suitable solvent until the liquid had nearly reached the top of the paper. Then the paper was removed from the solvent, dried and sprayed with a reagent that showed up the position of the spots of any metal ions present. The results are shown in the diagram. Which metal ions were present in solution S?

A only copper(II), iron(II) and iron(III) ions

B only copper(II), iron(II), iron(III) and nickel(II) ions

C iron(II), iron(III) and copper(II) ions with one other ion

D iron(II), iron(III), copper(II) and nickel(II) ions with one other ion

Correct Answer: A B C D

Explanation:

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